

LMS HANDLING SERVICES AVAILABLE by Annie

COMPLETE /HALF LMS HANDLING

FREE GUIDANCE

**ASSIGNMENTS QUIZ GDB CISCO ASSIGNMENTS WITH 90 -100
PERCENT GUARANTEED MARKS**

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Cs604 OPERATING SYSTEM LEC WISE
MCQS FROM LEC 1 TO 40

MOST IMP AND MOST REPEATED MCQS
BY ANNIE

Critical Section, Synchronization & Semaphores

1. Which of the following is **not** a requirement for solving the critical section problem?
 - A. Mutual Exclusion
 - B. Starvation
 - C. Progress
 - D. Bounded Waiting

✓ ☐ **Correct: B**
2. Peterson's algorithm is used for:
 - A. Deadlock prevention

- B. Memory management
- C. Process synchronization
- D. File protection

✓ ☐ **Correct: C**

3. Which instruction is commonly used in hardware solutions to implement mutual exclusion?
- A. ADD
 - B. LOAD
 - C. Test-and-Set
 - D. JMP
- ✓ ☐ **Correct: C**
4. What does the `wait()` operation in a semaphore do?
- A. Increments the semaphore
 - B. Decrements the semaphore
 - C. Terminates the process
 - D. Creates a process
- ✓ ☐ **Correct: B**
5. A binary semaphore can have values:
- A. 0 or 1
 - B. 0 to 9
 - C. True or False
 - D. Any real number
- ✓ ☐ **Correct: A**
6. Dining Philosophers Problem is used to demonstrate:
- A. File allocation
 - B. Deadlock
 - C. Memory leak
 - D. Paging
- ✓ ☐ **Correct: B**
7. A semaphore with an initial value of 1 is called:
- A. Counting semaphore
 - B. Mutex
 - C. Buffer
 - D. Thread
- ✓ ☐ **Correct: B**
8. Critical section problems arise in:
- A. Batch systems
 - B. Sequential processing
 - C. Multiprocessing
 - D. Spooling
- ✓ ☐ **Correct: C**
9. The `signal()` operation in a semaphore:
- A. Decreases the counter
 - B. Blocks the process
 - C. Increments the counter

D. Terminates the thread

✓ ☐ **Correct: C**

10. Which of the following is a classical problem of synchronization?

A. Process swapping

B. File indexing

C. Reader-Writer

D. Contiguous allocation

✓ ☐ **Correct: C**

11. Semaphore operations are performed using:

A. System calls

B. Interrupts

C. Signals

D. Libraries only

✓ ☐ **Correct: A**

12. Which mechanism allows multiple readers or a single writer?

A. Paging

B. Readers-Writers

C. Contiguous allocation

D. Threading

✓ ☐ **Correct: B**

13. The critical section can be exited using:

A. return()

B. exit()

C. signal()

D. release()

✓ ☐ **Correct: C**

14. Which of the following uses a flag array to indicate interest in entering the critical section?

A. Banker's Algorithm

B. Peterson's Algorithm

C. Linked List

D. Memory Table

✓ ☐ **Correct: B**

15. In a producer-consumer problem, the buffer is usually implemented using:

A. Arrays

B. Semaphore

C. Mutex only

D. Threads

✓ ☐ **Correct: B**

16. Busy waiting is a problem associated with:

A. Paging

B. Semaphore

C. Spooling

D. Context switching

✓ ☐ **Correct: B**

17. A process not in its critical section is in:
- A. Remainder section
 - B. Entry section
 - C. Kernel mode
 - D. Deadlock
- ✓ ☐ **Correct: A**
18. Mutual exclusion in software is harder because:
- A. Hardware is slow
 - B. Software is limited
 - C. No atomic operations
 - D. No hardware support
- ✓ ☐ **Correct: C**
19. Which of the following statements is true for deadlocks?
- A. All deadlocks can be avoided
 - B. All deadlocks can be detected
 - C. Deadlocks never occur
 - D. None of the above
- ✓ ☐ **Correct: B**
20. In semaphore terminology, P operation refers to:
- A. Wait
 - B. Signal
 - C. Sleep
 - D. Suspend
- ✓ ☐ **Correct: A**
21. Mutual exclusion can be ensured using:
- A. Delays
 - B. Signals
 - C. Critical sections
 - D. Buffering
- ✓ ☐ **Correct: C**
22. A flag in Peterson's algorithm is used to:
- A. Block interrupts
 - B. Indicate interest
 - C. Disable I/O
 - D. Allocate memory
- ✓ ☐ **Correct: B**
23. Which of these is a software solution to critical section?
- A. Test-and-Set
 - B. Interrupt disabling
 - C. Peterson's Algorithm
 - D. Preemption
- ✓ ☐ **Correct: C**
24. Which synchronization problem is solved using bounded buffer?
- A. Readers-Writers
 - B. Paging
 - C. Producer-Consumer

D. Disk Scheduling

✓ ☐ **Correct: C**

25. Semaphores are primarily used to handle:

A. File operations

B. Interrupts

C. Process synchronization

D. Memory leaks

✓ ☐ **Correct: C**

Deadlocks & Memory Management

26. A system is in a **deadlock state** if:

A. Processes terminate normally

B. Processes are waiting indefinitely

C. Resources are over-utilized

D. CPU utilization is maximum

✓ ☐ **Correct: B**

27. Which condition is **not** necessary for deadlock?

A. Circular wait

B. Preemption

C. Mutual exclusion

D. Hold and wait

✓ ☐ **Correct: B**

28. Which of the following helps avoid deadlock?

A. Round Robin

B. First-Come-First-Serve

C. Banker's Algorithm

D. FIFO

✓ ☐ **Correct: C**

29. Circular wait can be prevented by:

A. Assigning priorities to resources

B. Ignoring processes

C. Killing all processes

D. Using LRU

✓ ☐ **Correct: A**

30. Hold and Wait can be eliminated by:

A. Allowing processes to hold multiple resources

B. Making processes request all resources at once

C. Killing process

D. Allocating CPU first

✓ ☐ **Correct: B**

31. What is the state of a process holding a resource and waiting for another?

A. Running

B. Ready

C. Deadlocked

D. Suspended

✓ ☐ **Correct: C**

32. A deadlock-free system is:

A. Efficient

B. Safe

C. Non-preemptive

D. Inefficient

✓ ☐ **Correct: B**

33. In Banker's Algorithm, when can a process safely execute?

A. If available resources $\geq$ its need

B. If it is in the waiting state

C. If it's the oldest

D. If it requests no resources

✓ ☐ **Correct: A**

34. Deadlock detection involves:

A. Checking the process priority

B. Checking cyclic waits in graph

C. Checking processor time

D. Checking CPU scheduling

✓ ☐ **Correct: B**

35. Memory is divided into fixed-sized blocks in:

A. Segmentation

B. Paging

C. Stack

D. Cache

✓ ☐ **Correct: B**

36. In contiguous memory allocation, external fragmentation occurs when:

A. Processes terminate

B. Enough total memory is free but scattered

C. Memory is full

D. Paging is enabled

✓ ☐ **Correct: B**

37. Memory compaction is a solution for:

A. Thrashing

B. Page faults

C. External fragmentation

D. Stack overflow

✓ ☐ **Correct: C**

38. Which memory management scheme suffers from internal fragmentation?

A. Paging

B. Segmentation

C. Linked allocation

D. Swapping

✓ ☐ **Correct: A**

39. The page table contains:
A. Logical address
B. Process ID
C. Mapping of pages to frames
D. Threads
☒ **Correct: C**
40. Frame is a block in:
A. Logical memory
B. Virtual memory
C. Physical memory
D. File system
☒ **Correct: C**
41. Logical address = Page number + _____?
A. Offset
B. Frame
C. Size
D. Register
☒ **Correct: A**
42. Paging avoids:
A. Thrashing
B. Fragmentation
C. External fragmentation
D. Internal fragmentation
☒ **Correct: C**
43. The page size is:
A. Variable
B. Fixed
C. Zero
D. Infinite
☒ **Correct: B**
44. Which technique allows processes to be placed anywhere in memory?
A. Segmentation
B. Paging
C. Fragmentation
D. Synchronization
☒ **Correct: B**
45. Each process has its own:
A. Main memory
B. Page table
C. Kernel
D. I/O device
☒ **Correct: B**
46. The number of frames is determined by:
A. Page table
B. Logical memory
C. Physical memory

D. OS only

✓ ☐ **Correct: C**

47. Page tables are stored in:

A. Process address space

B. Main memory

C. CPU registers

D. Hard disk

✓ ☐ **Correct: B**

48. Which of the following does not suffer from fragmentation?

A. Contiguous allocation

B. Paging

C. Segmentation

D. Virtual memory

✓ ☐ **Correct: B**

49. Memory fragmentation that occurs outside allocated memory blocks is:

A. Internal

B. External

C. Physical

D. Logical

✓ ☐ **Correct: B**

50. When a process is removed from memory, the space is:

A. Compacted

B. Deleted

C. Deallocated

D. Locked

✓ ☐ **Correct: C**

Virtual Memory, Demand Paging & Page Replacement

51. Virtual memory is:

A. Memory on the hard disk

B. Part of cache

C. Execution of processes without full physical memory

D. An extension of main memory

✓ ☐ **Correct: C**

52. Virtual memory allows:

A. More users

B. Large virtual address space

C. Smaller programs

D. Disk scheduling

✓ ☐ **Correct: B**

53. Demand paging is:

A. All pages are loaded in advance

B. Pages loaded as needed

C. Only kernel pages are loaded

D. User requested paging

✓ ☐ **Correct: B**

54. When a required page is not in memory, it results in:

A. Page hit

B. Page replacement

C. Page fault

D. Segmentation fault

✓ ☐ **Correct: C**

55. Page fault occurs when:

A. Page is in memory

B. Page is not referenced

C. Page is not in memory

D. Memory is full

✓ ☐ **Correct: C**

56. Thrashing happens due to:

A. High CPU usage

B. Excessive swapping

C. No I/O

D. File fragmentation

✓ ☐ **Correct: B**

57. Which of the following avoids thrashing?

A. Increase number of processes

B. Reduce page size

C. Reduce degree of multiprogramming

D. Remove cache

✓ ☐ **Correct: C**

58. Which replacement algorithm removes the oldest page?

A. LRU

B. FIFO

C. Optimal

D. MRU

✓ ☐ **Correct: B**

59. LRU stands for:

A. Least Recently Used

B. Last Random User

C. Longest Ready Usage

D. Least Reserved Usage

✓ ☐ **Correct: A**

60. Optimal page replacement is based on:

A. Past use

B. Random choice

C. Future knowledge

D. Priority

✓ ☐ **Correct: C**

61. FIFO can cause:

- A. Page hit
- B. Belady's Anomaly
- C. Internal fragmentation
- D. Virtual fragmentation

✓ ☐ **Correct: B**

62. Belady's anomaly occurs in:

- A. Optimal algorithm
- B. FIFO algorithm
- C. LRU algorithm
- D. Clock algorithm

✓ ☐ **Correct: B**

63. LRU can be implemented using:

- A. Stack
- B. Queue
- C. Register
- D. Table

✓ ☐ **Correct: A**

64. When a page is brought into memory, it replaces:

- A. CPU
- B. Another page
- C. Disk
- D. Stack

✓ ☐ **Correct: B**

65. The working set model is used to:

- A. Calculate turnaround time
- B. Reduce thrashing
- C. Increase paging
- D. Increase memory

✓ ☐ **Correct: B**

66. Dirty bit indicates:

- A. Page is clean
- B. Page is shared
- C. Page has been modified
- D. Page is read-only

✓ ☐ **Correct: C**

67. In memory management, locality of reference implies:

- A. Memory is divided
- B. Recently used pages are used again
- C. Pages are swapped
- D. All memory is local

✓ ☐ **Correct: B**

68. Which of the following is a cause of page fault?

- A. Page already in memory
- B. Access to modified page
- C. Access to page not in memory

D. All pages are in RAM

✓ ☐ **Correct: C**

69. Copy-on-write is used in:

A. Swapping

B. Shared memory

C. Virtual memory

D. Page replacement

✓ ☐ **Correct: B**

70. Which page replacement algorithm is the most efficient?

A. FIFO

B. LRU

C. Optimal

D. Second Chance

✓ ☐ **Correct: C**

71. A system with more page faults is said to be:

A. Efficient

B. Thrashing

C. Optimized

D. Balanced

✓ ☐ **Correct: B**

72. When a program accesses instructions sequentially, it exhibits:

A. Temporal locality

B. Spatial locality

C. Page fault

D. Fragmentation

✓ ☐ **Correct: B**

73. A page table entry contains:

A. Address only

B. Page number only

C. Frame number + status bits

D. Logical address

✓ ☐ **Correct: C**

74. LRU uses a timestamp or counter to:

A. Track future use

B. Track last use

C. Track page size

D. Track process ID

✓ ☐ **Correct: B**

75. Increasing the number of frames sometimes:

A. Increases page faults

B. Decreases page faults

C. Has no effect

D. Reduces fragmentation only

✓ ☐ **Correct: A**

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